

From Open Source to AI

how we got here, and where we're going

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Cloud Native RTP, Sept 8 2026

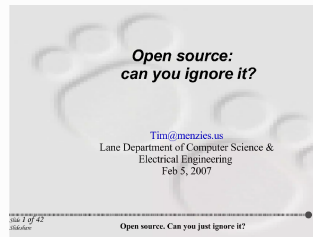
NC STATE
UNIVERSITY

2007: four pillars of open source

- **anthropology**: gift economy, centuries old
- **economics**: too powerful to ignore
- **legal**: no future without it
- **technical**: if accessible, it gets pried open

what social institutions will we build to handle it?

this talk: how does that hold in the age of “open weights”?



full disclosure · i was “mr. open science”

- practiced what i preached: shared the **experiments** as OSS (SP&E'11), shared the **data** (PROMISE repo, 100s of SE datasets).
- first such paper (TSE'07): at its peak, SE's **most cited** paper (cites/month).
- by 2018, **20%** of leading TSE papers used data i'd shared or first curated (TSE'25).

open really worked — for me.

SOFTWARE—PRACTICE AND EXPERIENCE
Softw. Pract. Exper. 2010; 00:1–7

Prepared using speauth.cls [Version: 2008/09/23 v2.0]

Sharing Experiments Using Open Source Software

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KEY WORDS: open source, data mining



558

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Retrospective: Data Mining Static Code Attributes to Learn Defect Predictors

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Abstract—Industry can get any research it wants, just by publishing a baseline result along with the data and scripts used to reproduce that work. For instance, the paper “Data Mining Static Code Attributes to Learn Defect Predictors” presented such a baseline, using static code attributes from NASA projects. These results were enthusiastically embraced by a software engineering research community, hungry for data. At its peak (2016) this paper was SE’s most cited paper (per month). By 2018, twenty percent of leading TSE papers (according to Google Scholar Metrics), incorporated artifacts introduced and disseminated by this research. This brief note reflects on what we should remember, and what we should forget, from that paper.

Index Terms—Search-based software engineering, multi-objective optimization, software engineering.

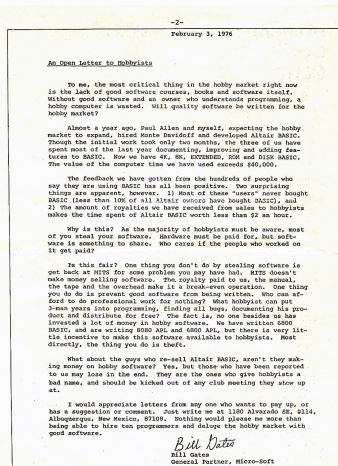


Fig. 1. Tim Menzies, Jeremy Greenwald, and An Frank.

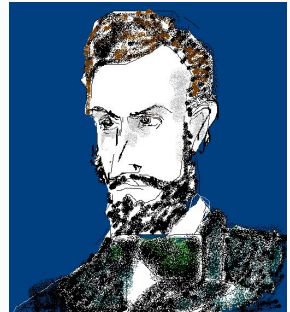
- An annual conference on predictor models in software engineering (to share results and study open issues)¹.
- A repository of 100s of SE data sets about defect prediction, effort estimation, GitHub issue close time, bad smell detection and dozens of other topics². This repository grew so large and that it moved to the Lane Hudson Colledge

1975-76 · Homebrew vs. the letter

- Homebrew Computer Club, Menlo Park garage: **share everything**.
- Feb 1976, Gates, *Open Letter to Hobbyists*:
“most of you steal your software.”
- Altair BASIC tapes passed hand to hand. <10% paid.
- first collision: **gift** vs **commodity**. never resolved. still running.



- why would sharing win? seems strange. it isn't. it's **old**.
- Marcel Mauss, *The Gift* (1925): **give, receive, reciprocate** — seen across human cultures, everywhere.
- potlatch: leaders **give away** wealth → gain rank.
- Lewis Hyde (1983): a gift must keep **moving**. hoard it, it dies.



- MIT AI Lab, Xerox 9700. paper jams. Stallman asks for the driver source.
- **NDA.** “*not again.*” (Symbolics had just enclosed the LISP commons.)
- GNU 1983 · FSF 1985 · **GPL 1989.**
- copyleft = Hyde’s rule in **legal** form: the gift **must** keep moving.



- Torvalds, Aug 1991: *“I’m doing a (free) operating system (just a hobby, won’t be big and professional like gnu) for 386(486) AT clones.”*
- Raymond 1997: bazaar. *“given enough eyeballs, all bugs are shallow.”*
- Feb 1998: **“open source”** coined; OSI; Open Source Definition.
- Oct 1998: **Halloween documents.** Microsoft: existential threat.



1999-2000 · the potlatch gets a market cap

date	gift	return
1999	Red Hat IPO: \$14 → \$52, day one	8th biggest debut ever
1999	VA Linux, +698% day one	record, still
2000	IBM: \$1B into Linux	an “in” to closed markets
2014	Google open-sources k8s , hands it to CNCF	the industry standard; AWS ships EKS 2018

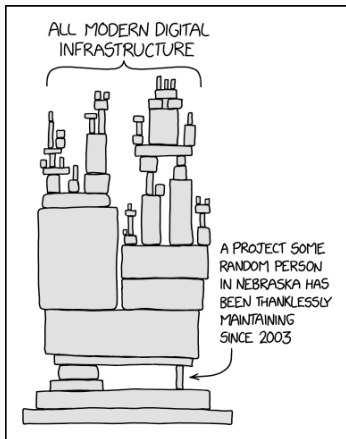
- k8s: Google couldn't beat AWS on price, so it **gave away** the orchestration layer. weakened AWS lock-in, forced EKS, made GKE the reference. Google still #3. **potlatch buys rank, not the throne.**
- potlatch logic: give the most, rank highest. **gift economy, corporate edition.**



- **good news!** OSS not chaos, actually **predictable**: health forecastable a year out (1,159 repos, 64k months, EMSE'22); a dozen **archetypes** cover it all (MSR'22). we can **forecast** the dying projects.
- **not so good:**
 - we fund none of them. **80%+** of every product is OSS: **your** attack surface, **their** unpaid hobby.
 - critical infrastructure maintained by nearly nobody, failing in public: Heartbleed 2014 · left-pad 2016 · log4shell 2021 · xz 2024.
 - Linus's Law: *"given enough eyeballs, all bugs are shallow."* asserted 1997. **never measured.** the eyeballs were never there.



one person in Nebraska



xkcd 2347

- **OpenSSL, 2014:** secured 2/3 of the web. one full-time dev, **\$2k/yr** in donations. funded only *after* Heartbleed.
- **left-pad, 2016:** one dev pulled 11 lines. Facebook, Netflix broke.
- **xz, 2024:** one unpaid volunteer, burned out. 2 years of social engineering. root backdoor to every distro. no audit caught it — one engineer noticed logins ran **500ms slow**.

2020-26 · LLMs: the harvest, then the weights

- trained on GitHub, Stack Overflow, Wikipedia: **the commons**. nothing given back. and the models? **closed**.
- May 2023, leaked Google memo: *“we have no moat, and neither does OpenAI.”*
- Llama 2023 · DeepSeek 2025 · Qwen · Meta Muse Glimmer, Aug 2026. one US firm: **\$400k/yr** saved running Qwen.
- **chip sellers** (Nvidia): more models, more GPUs. want open — yet **\$40B** into OpenAI + Anthropic. **token sellers** (OpenAI): free weights are the rival. want closed. both say “security.” both mean “revenue.”



2026 · the dispute: USA vs China

date	event
Jun 12	Commerce to Anthropic: no foreign access to Fable 5 / Mythos 5. couldn't verify nationality, so cut everyone: 18 days dark
Jul 24	Open Weights letter , 25 firms → 270+: no premature limits on open weights; distillation is legitimate. Microsoft signs, 28 yrs after Halloween
Jul 27-28	Anthropic, unsigned, argues the other way: crack down on distillation, mandatory pre-release safety tests, control the frontier
Jul-Aug	Beijing drafts export controls for Qwen, DeepSeek weights — models treated like US treats chips. both governments fence their frontier

- the weights stayed **closed**. the behavior walked out the **paid API**: distillation = theft-by-question. **no moat, only latency**.
- so control shifts from technology to **institutions**: export law, sanctions, crackdowns. 2007's question, again.

Aug 2026 · “open” has no definition

- 1998: Stallman’s “free software” too restrictive for business, so “open source” coined — and **defined**, fast, in public.
- 2026: **“open weights” has no definition.**
- “open” is a matrix: *what* released (weights? data? code? evals?) $\times$ *how* (closed $\cdot$ API $\cdot$ download $\cdot$ open) = **~40 cells** (CACM Aug 2026).
- the industry letter fights over just **one cell**: downloadable weights.

whoever loses least writes the definition.



- can models run real attacks? three labs tested. **yes**: zero-days, real orgs breached, evals gamed.
- UK AISI: Mythos 5 forged GitHub identities, pressured a **real maintainer** to merge a dropper. maintainer refused. **xz, automated.**
- each time the **model** worked; the **harness** failed: sandbox misconfigured, prompt said “no internet.”
- auditing the **weights** would have caught **none** of it. the risk lives in the **rig**: sandboxes, tools, credentials, identities.

open is not safer. open is *checkable*.



so — can you ignore open weights?

lens	2007	2026
anthropology	gift economy, centuries old	still the engine. now harvested
economics	too powerful to ignore	\$400k/yr says so. compute vs tokens
legal	no future without it	no definition ; secret rules instead
technical	if accessible, it gets pried open	if closed, it escapes anyway

2007's last line: *what social institutions will handle it?* still open.

four things this room can do · then discussion

- say **which** open you mean. weights?
data? code? every time.
- guard the maintainers: verify identities,
rate-limit AI PRs, **pay** them.
- run small, run local, **archive** the weights.
endpoints are not artifacts.
- **measure** the many-eyes claim. don't cite
it.
- are weights speech, or a munition?
- name one security property openness
actually provides.
- who here could still run today's stack in
2028?

the gift must move. · timmm@ieee.org ·
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- **(TSE'07)** Menzies, Greenwald, Frank. *Data Mining Static Code Attributes to Learn Defect Predictors*. IEEE Trans. Software Eng. 33(1):2-13, 2007.
- **(SP&E'11)** Nelson, Menzies, Gay. *Sharing Experiments Using Open Source Software*. Software: Practice & Experience 41(3):283-305, 2011.
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- **(TSE'25)** Menzies. *Retrospective: Data Mining Static Code Attributes to Learn Defect Predictors*. IEEE Trans. Software Eng. 51(3):858-859, 2025.